

EXPERMENT-13

13) Create an Interactive Scientific Calculator

HTML buttons and input fields, CSS Grid or Flexbox, JavaScript functions, arithmetic operators, Math object and click events.

CODE

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Scientific Calculator</title>

<style>

  * {

    box-sizing: border-box;

    margin: 0;

    padding: 0;

  }

  body {

    font-family: Arial, sans-serif;

    background: linear-gradient(135deg, #141e30, #243b55);

    min-height: 100vh;

    display: flex;

    justify-content: center;

    align-items: center;

  }
```

```
.calculator {  
    width: 360px;  
    background: #1e1e1e;  
    padding: 20px;  
    border-radius: 20px;  
    box-shadow: 0 15px 35px rgba(0,0,0,0.5);  
}
```

```
h2 {  
    color: white;  
    text-align: center;  
    margin-bottom: 15px;  
}
```

```
#display {  
    width: 100%;  
    height: 70px;  
    border: none;  
    border-radius: 10px;  
    background: #111;  
    color: #00ffcc;  
    font-size: 25px;  
    text-align: right;  
    padding: 10px;  
    margin-bottom: 15px;  
}
```

```
.buttons {
```

```
display: grid;
grid-template-columns: repeat(4, 1fr);
gap: 10px;
}
```

```
button {
  height: 55px;
  border: none;
  border-radius: 10px;
  font-size: 17px;
  cursor: pointer;
  background: #333;
  color: white;
}
```

```
button:hover {
  background: #555;
}
```

```
.scientific {
  background: #4b3f72;
}
```

```
.operator {
  background: #ff8c42;
}
```

```
.clear {
```

```
        background: #e63946;
    }

    .equal {
        background: #00a896;
    }

    @media (max-width: 400px) {
        .calculator {
            width: 95%;
        }
    }
</style>
</head>

<body>

<div class="calculator">

    <h2>Scientific Calculator</h2>

    <input type="text" id="display" readonly>

    <div class="buttons">

        <button class="clear" onclick="clearDisplay()">C</button>

        <button class="scientific" onclick="calculate('sqrt')">√</button>

        <button class="scientific" onclick="calculate('square')">x²</button>

        <button class="operator" onclick="append('/')">÷</button>
```

<button class="scientific" onclick="calculate('sin')">sin</button>
<button class="scientific" onclick="calculate('cos')">cos</button>
<button class="scientific" onclick="calculate('tan')">tan</button>
<button class="operator" onclick="append('*')">x</button>

<button class="scientific" onclick="calculate('log')">log</button>
<button class="scientific" onclick="calculate('ln')">ln</button>
<button class="scientific" onclick="calculate('pi')"> π </button>
<button class="operator" onclick="append('-')">-</button>

<button onclick="append('7')">7</button>
<button onclick="append('8')">8</button>
<button onclick="append('9')">9</button>
<button class="operator" onclick="append('+')">+</button>

<button onclick="append('4')">4</button>
<button onclick="append('5')">5</button>
<button onclick="append('6')">6</button>
<button class="scientific" onclick="calculate('factorial')">n!</button>

<button onclick="append('1')">1</button>
<button onclick="append('2')">2</button>
<button onclick="append('3')">3</button>
<button class="scientific" onclick="calculate('inverse')">1/x</button>

<button onclick="append('0')">0</button>
<button onclick="append('.')">.</button>
<button class="equal" onclick="calculate('equal')">=</button>

```
    <button class="scientific" onclick="append('*')">xʸ</button>
  </div>
</div>
```

```
<script>

  const display = document.getElementById("display");

  function append(value) {
    display.value += value;
  }

  function clearDisplay() {
    display.value = "";
  }

  function calculate(type) {
    try {
      let value = Number(display.value);

      if (type === "equal") {
        display.value = eval(display.value);
      }

      else if (type === "sqrt") {
        display.value = Math.sqrt(value);
      }

      else if (type === "square") {
```

```
    display.value = Math.pow(value, 2);  
}
```

```
else if (type === "sin") {  
    display.value = Math.sin(value * Math.PI / 180);  
}
```

```
else if (type === "cos") {  
    display.value = Math.cos(value * Math.PI / 180);  
}
```

```
else if (type === "tan") {  
    display.value = Math.tan(value * Math.PI / 180);  
}
```

```
else if (type === "log") {  
    display.value = Math.log10(value);  
}
```

```
else if (type === "ln") {  
    display.value = Math.log(value);  
}
```

```
else if (type === "pi") {  
    display.value = Math.PI;  
}
```

```
else if (type === "inverse") {
```

```
        display.value = 1 / value;
    }

    else if (type === "factorial") {
        let result = 1;
        for (let i = 1; i <= value; i++) {
            result *= i;
        }
        display.value = result;
    }
}
catch {
    display.value = "Error";
}
}
```

</script>

</body>

</html>

OUTPUT

